7.2 Route of Entry

To establish an infection, pathogen must first enter the host. Normal defense mechanisms and barriers (For example Skin, mucus, ciliated epithelium, lysozyme) make it difficult for the pathogen to enter the body.

Sometimes these barriers are break through for example cut in the skin, wound, tumor, ulcer which provides portal of entry for the bacteria. Some bacterial pathogens have the means to overcome the barriers through various virulence factors and invade the body.

Certain bacteria are infective when introduced through optimal route. The various route of entry of pathogens, which are cut or abrasion or wound (skin), Ingestion, Inhalation, arthropod bite, sexual transmission and congenital transmission. These are already explained in the XI Standard text book. The various bacterial pathogens, its pathogenesis clinical symptoms, laboratory diagnosis, control, prophylaxis and treatment with appropriate antibiotics are discussed below.

7.3 *Staphylococcus Aureus* (Pyogenic Cocci)

The genus *Staphylococcus* is included in the family Micrococcaceae. *Staphylococcus* is a normal flora of skin and mucous membranes, but it accounts for human infections, which is known as staph infection. The name *Staphylococcus* was derived from a Greek word, **'staphyle'** means **bunch of grapes** and **'kokkos'** means **berry**. *Staphylococcus aureus* is a pathogenic species that causes pyogenic infections in human.

7.3.1 Morphology

- *Staphylococci* are gram positive spherical cocci, (0.8µm–1.0µm in diameter) arranged characteristically in grape like clusters (Figure 7.1).
- They are non-motile and non-sporing and few strains are capsulated.



The grape like cluster formation in *Staphylococcus aureus* is due to cell division occurring in three perpendicular planes, with daughter cells tending to be remaining in close proximity.

7.3.2 Cultural Characteristics

- They are aerobes and facultative anaerobes, optimal temperature is 37°C and optimum pH is 7.4–7.6.
- They grow on the following media and shows the characteristic colony morphology (Table 7.1 & Figure 7.2).

Table 7.1: Staphylococci aureus colony morphology on various media

Media	Colony Morphology
Nutrient Agar	Colonies are circular, smooth, convex, opaque and produces golden yellow pigment (most strains).
Blood Agar	Beta haemolysis
Mannitol salt Agar (MSA)	It is a selective medium for S. *aureus* produces yellow colored colonies due to fermentation of mannitol

TN_GOVT_XII_Micro_Biology_CH07.indd 115 15-04-2019 16:38:23